

● *Original Contribution*

ULTRASOUND-BASED MEASUREMENT OF MOLECULAR MARKER CONCENTRATION IN LARGE BLOOD VESSELS: A FEASIBILITY STUDY

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Abstract—Ultrasound molecular imaging has demonstrated efficacy in pre-clinical studies for cancer and cardiovascular inflammation. However, these techniques often require lengthy protocols because of waiting periods or additional control microbubble injections. Moreover, they are not capable of quantifying molecular marker concentration in human tissue environments that exhibit variable attenuation and propagation path lengths. Our group recently investigated a modulated acoustic radiation force-based imaging sequence, which was found to detect targeted adhesion independent of control measurements. In the present study, this sequence was tested against various experimental parameters to determine its feasibility for quantitative measurements of molecular marker concentration. Results indicated that measurements obtained from the sequence (residual-to-saturation ratio, R_{resid}) were independent of acoustic pressure and attenuation ($p > 0.13$, $n = 10$) when acoustic pressures were sufficiently low. The R_{resid} parameter exhibited a linear relationship with measured molecular marker concentration ($R^2 > 0.94$). Consequently, feasibility was illustrated *in vitro*, for quantification of molecular marker concentration in large vessels using a modulated acoustic radiation force-based sequence. Moreover, these measurements were independent of absolute acoustic reflection amplitude and used short imaging protocols (3 min) without control measurements. (E-mail: jh7fj@virginia.edu) © 2015 World Federation for Ultrasound in Medicine & Biology.

Key Words: Ultrasound, Molecular imaging, Quantitative measurement, Acoustic radiation force.

INTRODUCTION

Microbubbles typically comprise of micrometer-sized low solubility gas core stabilized with a shell that acts as a diffusion barrier. Thanks to their high reflectivity of acoustic energy, microbubbles are the predominant ultrasound contrast agent. In addition to their use as a blood tracer, they can also be used for enhanced molecular imaging and drug/gene delivery (De Jong et al., 1992; Deshpande et al., 2010; Ferrara et al., 2007; Klibanov, 2002; Lindner, 2004; Mayer and Bekerredjian, 2008). In ultrasound-based targeted molecular imaging, ligands are matched to molecular markers associated with the disease of interest (e.g., ICAM-1 and p-selectin for cardiovascular disease) and are conjugated to the microbubble shell. Consequently, the specific ligand-receptor bond allows targeted microbubbles to adhere to specific regions

of the vascular endothelium, providing an increased ultrasound echo signal indicating the presence of specific molecular markers on the vessel wall (Kornmann et al., 2010; Lindner et al., 2001; Villanueva et al., 1998; Weller et al., 2002). To increase adhesion of targeted microbubbles to the vessel wall, multiple ligands can be incorporated on the microbubble shell (Ferrante et al., 2009). Acoustic radiation force (ARF) is frequently applied to increase binding efficiency of microbubbles to potential sites, for example, in large vessel environments with high flow velocity and shear force (Dayton et al., 1999; Gessner et al., 2012; Rychak et al., 2007; Zhao et al., 2004). At present, ultrasound molecular imaging techniques are still mainly limited to pre-clinical applications.

The detection and isolation of echo signal derived from the molecularly bound microbubbles (adherent microbubbles) are essential in ultrasound-based targeted molecular imaging. Signals from surrounding tissue and freely circulating microbubbles must be suppressed to prevent false positive detection of adherent

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microbubbles. Typically, the combination of non-linear signal detection (e.g., pulse inversion or harmonic imaging) and lengthy waiting periods (e.g., 15–30 min) is used to isolate signals from adherent microbubbles (Deshpande *et al.*, 2010; Lindner *et al.*, 2001; Phillips and Gardner, 2004; Phillips, 2001; Pochon *et al.*, 2010). To decrease the procedure time, several “fast-imaging” methods (Hu *et al.*, 2013; Pysz *et al.* 2012a, 2012b) and real-time imaging methods (Zhao *et al.*, 2007; Gessner *et al.*, 2010; Hu *et al.*, 2010; Needles *et al.*, 2009) have been developed for small vessel environments. Low-pass inter-frame filtering or pre- and post-burst subtraction are often used in these methods to separate signals of adherent microbubbles from free circulating microbubbles. In addition, with the assistance of ARF, recent studies have also observed ultrasound targeted molecular imaging in large vessel environments using some new techniques (Mauldin *et al.*, 2012; Patil *et al.*, 2011; Wang *et al.* 2013b). In summary, state-of-the-art ultrasound molecular imaging methods have demonstrated efficacy in pre-clinical studies for detection and monitoring of various types of cancers (Bzyl *et al.*, 2011; Fischer *et al.*, 2010; Pysz *et al.*, 2010) as well as cardiovascular inflammation (Kaufmann *et al.*, 2007; Masseau *et al.*, 2012). However, these approaches possess some limitations such as requiring a “control injection” (i.e., separate injection with non-targeted microbubbles) to estimate the non-specific “background” signal. Consequently, the entire procedure times are still quite long (up to 30–40 min) because of injection of multiple microbubble populations and waiting periods needed between consecutive injections (Lindner *et al.*, 2000; Masseau *et al.*, 2012; Pysz *et al.* 2012b). This characteristic is undesirable because lengthy or complex imaging protocols could delay clinical adoption and also directly, or indirectly, increase procedure costs. Additionally, none of these methods have proven capable to quantify the molecular marker concentration in human tissue environments where attenuation and imaging path lengths are highly variable. The ability to achieve quantitative measurements of molecular marker concentration with non-invasive, low-cost ultrasound holds value in a number of possible clinical applications. For example, quantitative measurements could facilitate time serial monitoring of disease or therapy at the molecular level or enable the establishment of measurement thresholds as a basis for treatment or interventions.

To meet the desired goals of quantitative molecular marker measurements with fast, simple imaging protocols, our group recently investigated a new modulated ARF-based imaging sequence. This strategy uses relatively short imaging protocols (3 min) that did not require lengthy waiting periods (Wang *et al.* 2013a). Briefly, this sequence provided constant ARF for a given period and

collected acoustic data of adherent microbubbles continuously, both with application and after cessation of ARF. Parameters related to the acoustic response of microbubbles were extracted and used to quantify the difference between specific and non-specific adhesion of microbubbles. Although quantitative assessment of disease progression has been reported in small blood vessel environment (Bachawal *et al.*, 2013; Pysz *et al.* 2012a, 2012b), these methods relied on absolute signal magnitude, which assumes consistent attenuation and imaging path lengths to the disease site, which are highly variable in human imaging environments. Additionally, groups have used percent contrast coverage area as a quantitative measurement (Pysz *et al.* 2012a, 2012b). This approach attempts to quantify the percentage of pixels in the target region that contain adherent microbubbles. Although found to be effective in cancers environments as a quantitative measurement proportional to microvessel density (Pysz *et al.* 2012a, 2012b), it does not allow for measurements of actual marker concentration. In this paper, we report that it is feasible to achieve quantitative measurements of molecular marker concentration in large vessel environments by using a modulated ARF-based imaging sequence. The effects of acoustic pressures, intensities, and attenuation on the modulated ARF-based imaging sequence are studied. The measured concentrations of molecular markers are correlated with parameter extracted from the imaging sequence.

METHODS

In vitro experiment setup

Large blood vessel environments were mimicked using flow phantoms as described in previous studies (Fig. 1) (Hall *et al.*, 1997; Wang *et al.*, 2012). Briefly, the liquid channels were molded using borosilicate glass rods (McMaster-Carr, Robbinsville, NJ, USA) with a diameter of 4.0 mm, which were placed horizontally inside a custom designed phantom holder (Applied Rapid Technologies, Fredericksburg, VA,

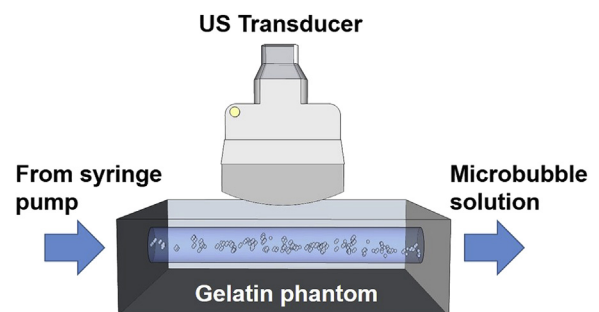


Fig. 1. *In vitro* flow phantom setup. US = ultrasound.

USA). The mixture of gelatin (6.1%, w/w, Type B, Fisher Scientific, Pittsburgh, PA, USA), agar (2.2%, w/w, Acros Organics, Geel, Belgium) and graphite (1.8%, w/w, < 20 μm , Sigma-Aldrich, St. Louis, MO, USA) were dissolved in boiling water and then poured into the phantom holder with glass rods at a temperature of approximately 75 °C. The glass rods were slowly removed after the phantom was allowed to cool to room temperature. The thickness of the phantom was kept constant at approximately 8 mm from the top vessel wall to the phantom surface. All liquid channels were determined to possess a smooth, specular reflecting inner surface using both optical and acoustic assessments.

Microbubbles used in this study were synthesized biotinylated lipid shelled microbubbles (mean diameter $\approx$ 2.2 μm , standard deviation of diameter $\approx$ 1.5 μm , median diameter $\approx$ 1.9 μm) (Klibanov, 2002; Klibanov et al., 1999). Empirically, it is preferable to use ultrasound pulses for pushing these microbubbles at a transmit frequency of approximate 4 MHz (Patil et al., 2011). Microbubbles were diluted in 0.9% Sodium Chloride Irrigation, USP (Baxter Healthcare, Deerfield, IL, USA) at a concentration of $0.5 \times 10^6/\text{mL}$. The concentration of microbubbles in this aqueous solution was measured and maintained by a Coulter counter (Coulter Multisizer 3, Beckman Coulter, Brea, CA, USA). Fresh microbubble dispersion was prepared every 10 min during the experiments. In addition, 0.5% (w/w) bovine serum albumin (BSA, Sigma-Aldrich, St. Louis, MO, USA) solution was added into the microbubble solution to further prevent non-specific adhesion (Hernot et al., 2012; Wang et al. 2013c).

Two types of phantom channels were used in the study: targeted phantom channels and control phantom channels. Targeted phantom channels were created by incubation with streptavidin (AnaSpec, Fremont, CA, USA) solution (concentration ranging from 1 to 50 $\mu\text{g}/\text{mL}$) for 12 h (4°C) followed by another incubation with 10% (w/w) BSA solution for 10 min (4°C). Control phantom channels were created by incubation with 10% (w/w) BSA solution alone for 12 h (4°C). Each phantom channel was used for one trial only to avoid effects from residual microbubbles (Wang et al. 2013c).

Experiments were performed by injecting microbubble solution through phantom channels at a given flow velocity (6–10 cm/s) using a programmable syringe pump (PHD ULTRA, Harvard Apparatus, Holliston, MA, USA) for 180 s. The proposed phantom channel diameter and flow velocities were similar to the typical diameter and average flow velocities in large human vessels, such as in the brachial artery (4 mm and 9 cm/s) (Ku, 1997; Levenson et al., 2000; Safar et al., 1981).

Modulated ARF-based imaging sequence

Similar to a previous study by our group (Wang et al. 2013a), the modulated ARF-based imaging sequence continuously collected acoustic echo data for 180 s at a frame rate of 4 Hz (Fig. 2). The sequence was divided into three sections: 10 s of imaging alone (0–10 s), 70 s of imaging and ARF (10–80 s) and 100 s of imaging alone (80–180 s). The imaging pulses were single cycle plane wave pulses at a center frequency of 4.5 MHz. Echo data obtained from three consecutive imaging pulses were used to form one image frame using conventional dynamic receive focusing. All imaging pulses for the entire 180 s sequence were identical, and were neither amplitude nor phase modulated. The peak negative pressure (PNP) of imaging pulses in this study ranged from 17.3 to 91.6 kPa, measured using a calibrated Onda GL-0200 hydrophone (Onda, Sunnyvale, CA, USA), with corresponding mechanical indices (MI) ranging from 0.008 to 0.043. In addition, the ARF pulses were 15-cycle plane wave pulses at a center frequency of 4.5 MHz. The average pulse repetition frequency (PRF) of ARF pulses ranged from 5 to 15 kHz. And the PNP of ARF pulses ranged from 46.5 to 189.5 kPa, with corresponding MI ranging from 0.022 to 0.089. It was confirmed by fluorescent microscopy in previous studies that adherent microbubbles would accumulate along the bottom liquid channels with application of ARF pulses (Mauldin et al., 2012; Patil et al., 2011). The above imaging sequence was programmed on a Verasonics programmable ultrasound scanner (Verasonics, Redmond, WA, USA) equipped with an ATL L12-5 38 mm linear array transducer (Philips Healthcare, Andover, MA, USA).

For each trial, 720 consecutive image frames were processed using MATLAB (Mathworks, Natick, MA, USA). For each image frame, the average signal magnitude (M) was calculated within the region of interest (ROI) (Wang et al. 2013a), which was located in the center of the bottom channel wall and had a depth of 0.2 mm and a width of 2 mm. The depth location of the ROI was determined automatically by detecting the location of maximum average signal magnitude. The ROI also spanned 10 consecutive laterally adjacent A-lines to decrease the effects of non-uniform distribution of adherent microbubbles along the bottom channel wall. The signal magnitude curve ($M(t)$, $0 \leq t \leq 180$ s) was obtained by computing M for all 720 image frames.

It has been reported (Wang et al. 2013a) that $M(t)$ for targeted phantom channels expresses an exponential increase approaching a saturation magnitude during the second section of the imaging sequence with application of ARF (10–80 s); and an exponential decay approaching a residual magnitude during the third section of the imaging sequence after cessation of ARF (80–180 s). The ratio

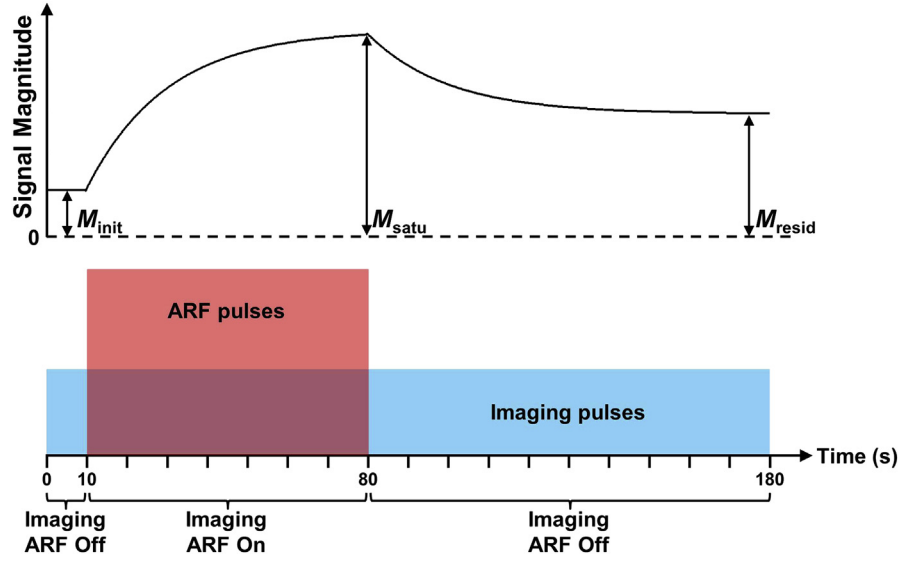


Fig. 2. Diagram of the modulated ARF-based imaging sequence (bottom) and corresponding representative signal magnitude curve (top). The signal magnitude curve is acquired in a region of interest corresponding to the lower vessel wall where radiation force leads to the aggregation of adherent microbubbles. Imaging and ARF pulses are represented with blue and red colors, respectively. The amplitudes of imaging and ARF pulses are not to scale.

of residual to saturation signals (R_{resid}) was used to separate targeted adhesion from non-specific adhesion:

$$R_{\text{resid}} = \left(\frac{M_{\text{resid}} - M_{\text{init}}}{M_{\text{satu}} - M_{\text{init}}} \right) \times 100\% \quad (1)$$

M_{init} and M_{resid} were the initial signal magnitude (signal magnitude at $t = 5$ s) and residual signal magnitude (signal magnitude at $t = 180$ s), respectively (Fig. 2). M_{satu} was the saturated signal magnitude (signal magnitude at $t = 80$ s for targeted channels). In this study, R_{resid} was calculated and used to compare targeted adhesion levels in different experimental conditions. According to the models described in a previous study by our group (Wang *et al.* 2013a), M_{satu} represented the full saturation of adherent microbubbles (both specific and non-specific) and M_{init} represented the background signal magnitude of the wall in the absence of adherent microbubbles. After cessation of ARF, only targeted ligand–receptor bound microbubbles remained and the remaining adherent microbubble concentration is quantified by M_{resid} . The R_{resid} parameter combines the signal magnitude-dependent parameters (M_{init} , M_{satu} , and M_{resid}) to produce a measure that is independent of absolute signal magnitude. It represents the percentage of wall containing targeted ligand–receptor bound microbubbles. Consequently, in this study we hypothesize that R_{resid} has a linear relationship with molecular marker concentration along the vessel wall and that R_{resid} is independent of attenuation or imaging path length effects on acoustic signal magnitude.

Measurements of molecular marker concentration

To estimate the concentration of streptavidin bound to the phantom channels at different incubation concentrations (1–50 $\mu\text{g/mL}$), Eu^{3+} -labelled streptavidin (DEL-FIA Eu-N1 Streptavidin, PerkinElmer, Waltham, MA, USA) was used to incubate phantom channels at corresponding incubation concentrations (1–50 $\mu\text{g/mL}$) using exactly the same procedure as for regular streptavidin: 12 h incubation followed by another incubation with 10% (w/w) BSA solution for 10 min. One-centimeter-long phantom channels were harvested and then the inner channel surfaces were washed for 15 s using 100 μL Enhancement Solution (DEL-FIA, PerkinElmer, Waltham, MA, USA). The enhancement solution was then collected; and the corresponding Eu^{3+} -fluorescence was measured in a dissociation-enhanced time-resolved spectrofluorometry (SPECTRAmax GEMINI-XS, Molecular Devices, Sunnyvale, CA, USA). Finally, the concentration of Eu^{3+} -labelled streptavidin in the samples was computed by comparing with the fluorescence signal of a reference solution. The measurements were repeated 5 times for each incubation concentration of Eu^{3+} -labelled streptavidin (Suonpää *et al.*, 1992).

Experiment design

The first set of experiments was designed to test the effects of acoustic pressures of both imaging and ARF pulses on the parameter R_{resid} . The acoustic parameters (PNP and MI) of different groups (from G_{P1} to G_{P9}) are listed in Table 1. For all 9 groups, the following

Table 1. Parameters of experiments to test effects of acoustic pressures

Parameter	Group name								
	G _{P1}	G _{P2}	G _{P3}	G _{P4}	G _{P5}	G _{P6}	G _{P7}	G _{P8}	G _{P9}
PNP _{imaging} * (kPa)	17.3	17.3	17.3	46.1	46.1	46.1	91.6	91.6	91.6
MI _{imaging} †	0.008	0.008	0.008	0.022	0.022	0.022	0.043	0.043	0.043
PNP _{ARF} ‡ (kPa)	46.5	106.0	189.5	46.5	106.0	189.5	46.5	106.0	189.5
MI _{ARF} §	0.022	0.050	0.089	0.022	0.050	0.089	0.022	0.050	0.089

* Peak negative pressure of imaging pulses.

† Mechanical index of imaging pulses.

‡ Peak negative pressure of acoustic radiation force (ARF) pulses.

§ Mechanical index of acoustic radiation force (ARF) pulses.

parameters remained the same: flow velocity (6 cm/s), incubation concentration of streptavidin (50 $\mu\text{g/mL}$), and PRF of ARF pulses (5 kHz). For each group, 10 trials of targeted phantom channels and 10 trials of control phantom channels were performed in different phantoms to eliminate measurement bias from phantom-to-phantom or batch-to-batch microbubble variations. The R_{resid} parameter of each group was calculated and compared. For Student *t*-tests, differences were considered statistically significant only if the calculated *p*-value was less than 0.05. Based on the results, the group with best performance (details described below in results) was selected as the “reference group” and the acoustic parameters of the reference group were used for the next set of experiments.

The next set of experiments was designed to test the effect of attenuation, and thus absolute signal magnitude, on the parameter R_{resid} . The flow and acoustic parameters of the aforementioned reference group were implemented. One layer of 0.8-mm-thick rubber membrane (Neoprene Rubber, McMaster–Carr, Robbinsville, NJ, USA) was placed between the transducer and phantom for the first group, G_{A1}. For the second group, G_{A2}, two layers of the same rubber membrane were placed between the transducer and phantom. The measured acoustic parameters of PNP and MI of the two groups are listed in Table 2. Based on the measurements, one and two layers of 0.8-mm-thick rubber membrane provided 2.1 dB and

4.4 dB attenuation relative to the reference group (G_{P5}), respectively. For each group, 10 trials of targeted phantom channels and 10 trials of control phantom channels were performed.

The third set of experiments was designed to test the effect of ARF pulse intensities (time-averaged intensity of ARF pulses) on the parameter R_{resid} . Our previous study reported the effect of lower PRF of ARF pulses (0.5 and 2.5 kHz) (Wang et al. 2013a). In this study, two groups (G_{PRF1} and G_{PRF2}) of experiments with higher PRF of ARF pulses (10 and 15 kHz) were performed. Except for ARF pulse intensity, other flow and acoustic parameters used in this part of experiments were the same as the aforementioned reference group. For each group, 10 trials of targeted phantom channels and 10 trials of control phantom channels were tested.

The fourth set of experiments was designed to test the ability of this technique to quantitatively measure molecular marker concentration based on the parameter R_{resid} . The flow and acoustic parameters of different groups of experiments (G_{V1 C1} to G_{V2 C6}) are listed in Table 2. For the first 6 groups (G_{V1 C1}, G_{V1 C2}, G_{V1 C3}, G_{V1 C4}, G_{V1 C5}, G_{V1 C6}), the flow velocity remained the same at 6 cm/s and incubation concentration of streptavidin increased from 1 to 50 $\mu\text{g/mL}$. For the second 6 groups (G_{V2 C1}, G_{V2 C2}, G_{V2 C3}, G_{V2 C4}, G_{V2 C5}, G_{V2 C6}), flow velocity remained the same at a higher level, 10 cm/s. All other flow and acoustic parameters used in these 12 groups of experiments, except for flow velocity and incubation concentration of streptavidin, were kept the same as the aforementioned reference group. For each group, 10 trials of targeted phantom channels and 10 trials of control phantom channels were tested.

RESULTS

Effect of varying acoustic pressures of imaging and ARF pulses on R_{resid}

The signal magnitude curves of both targeted and control phantom channels for different acoustic conditions listed in Table 1 (G_{P1} to G_{P9}) are illustrated in

Table 2. Parameters of experiments to test effects of attenuation

Parameter	Group name	
	G _{A1} *	G _{A2} †
PNP _{imaging} (kPa)	28.9	25.3
MI _{imaging}	0.014	0.013
PNP _{ARF} (kPa)	83.1	63.6
MI _{ARF}	0.041	0.031

* Attenuation group with 2.1 dB attenuation relative to reference group (G_{P5}).† Attenuation group with 4.4 dB attenuation relative to reference group (G_{P5}).

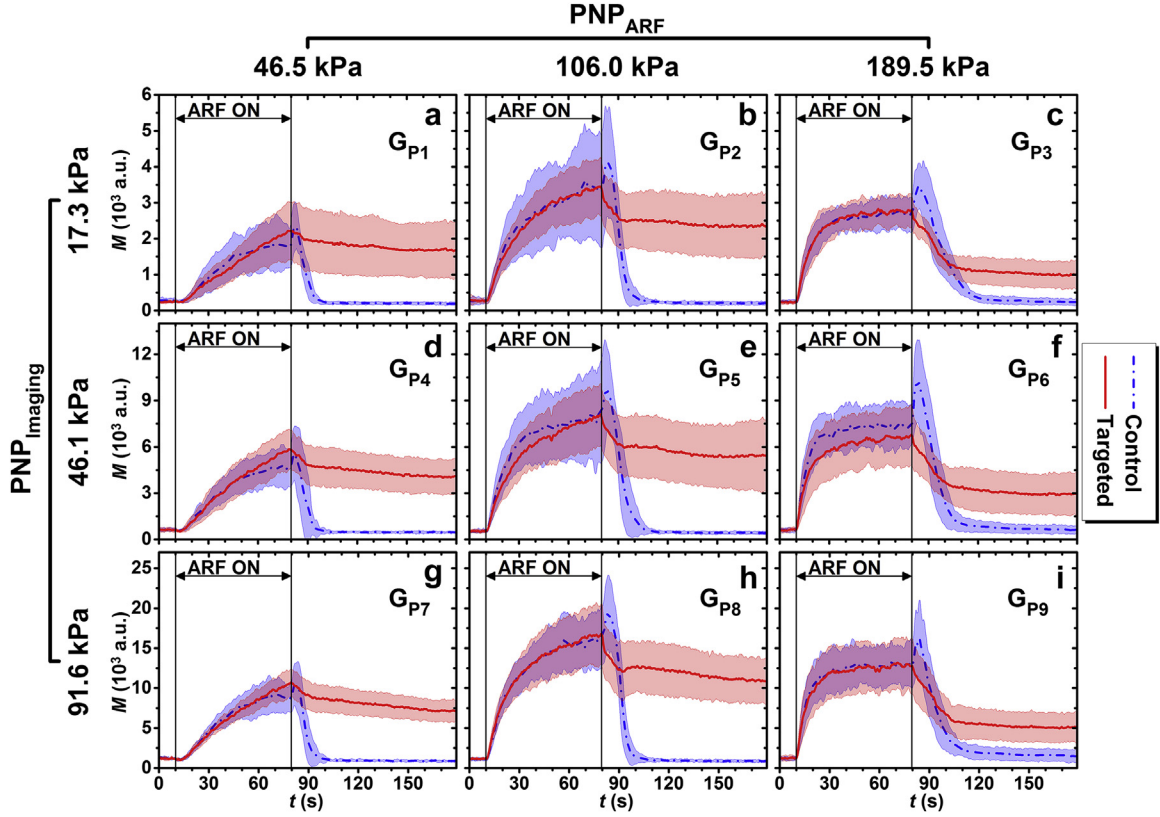


Fig. 3. Signal magnitude curves in regions of adherent microbubbles in control (blue) and targeted (red) phantom channels under different acoustic conditions listed in Table 1. (a) G_{P1} , (b) G_{P2} , (c) G_{P3} , (d) G_{P4} , (e) G_{P5} , (f) G_{P6} , (g) G_{P7} , (h) G_{P8} , (i) G_{P9} . Blue dash-dot and red solid lines indicate the mean values from 10 trials. Light color shadows indicate the corresponding error bars at the range of [mean $\pm$ standard deviation]. Different y-axis scales used to make comparison between curve shapes easier.

Figure 3. Each curve represents the average over 10 trials. The signal magnitude curves of both targeted and control phantom channels for all 9 groups increased with application of ARF (10–80 s). After cessation of ARF (80–180 s), signal magnitude returned to the baseline level for control phantom channels; and maintained a significantly higher residual level for targeted phantom channels ($p < 0.001$, $n = 10$). The R_{resid} parameter for both targeted and control phantom channels across various acoustic conditions listed in Table 1 are illustrated in Figure 4. The mean R_{resid} values were below 5% for control phantom channels at different groups; and were above 25% for targeted phantom channels at different groups. In addition, for each group, the R_{resid} of targeted phantom channels was significantly higher than that of the control phantom channels ($p < 0.001$, $n = 10$).

At any given imaging pulse acoustic pressure (17.3, 46.1, or 91.6 kPa), the median acoustic pressure (106.0 kPa) of ARF pulses tested in the study resulted in both the highest saturation and residual average signal magnitude for targeted phantom channels. In addition, the highest acoustic pressure (189.5 kPa) of ARF pulses

resulted in the lowest residual signal magnitudes of targeted phantom channels. Similarly, the three groups (G_{P3} , G_{P6} , and G_{P9}) with application of the highest ARF pulse pressure expressed a relatively lower R_{resid}

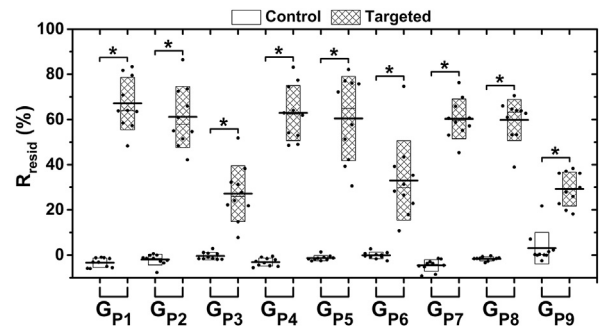


Fig. 4. The R_{resid} parameter across the different acoustic conditions listed in Table 1. Boxes (empty for control and filled for targeted phantom channels) show the range of [mean $\pm$ standard deviation]. Black lines located at the center of the boxes show the corresponding mean value from 10 trials. Raw data from 10 trials are shown as solid dots overlaid with corresponding boxes. For Student's t -test, *: $p < 0.001$, $n = 10$.

value of approximate 30%. There was no significant difference between any two of these three groups ($p > 0.40$, $n = 10$). In addition, other groups (G_{P1} , G_{P2} , G_{P4} , G_{P5} , G_{P7} , and G_{P8}) expressed a relatively higher R_{resid} value of approximate 60%, with no significant difference between any two of the 6 groups ($p > 0.13$, $n = 10$).

At a given ARF pulse pressure (46.5, 106.0, or 189.5 kPa), higher imaging pulse pressure resulted in higher signal magnitude for both targeted and control phantom channels. However, the shape of the average signal magnitude curves (both targeted and control phantom channels) remained similar for all three imaging pulse pressures (17.3, 46.1, 91.6 kPa). With regards to the R_{resid} parameter in targeted phantom channels, at a given ARF pulse pressure, there was no significant difference in R_{resid} values between any two of the three groups at different imaging pulse pressure ($p > 0.16$ for PNP_{ARF} of 46.5 kPa, $p > 0.80$ for PNP_{ARF} of 106.0 kPa, and $p > 0.40$ for PNP_{ARF} of 189.5 kPa, $n = 10$).

Based on the results in Figures 3 and 4, G_{P5} was selected as the reference group for the following reasons: (1) the G_{P5} group was among the groupings with highest R_{resid} ($\sim 60\%$); (2) the voltage profile applied for both imaging and ARF pulses were the same (3 V) which meant that there were no possible errors derived from switching the power supply between high voltage and low voltage profiles. The acoustic and flow parameters of the reference group were used for the following sets of experiments.

Effects of attenuation on R_{resid}

The signal magnitude curves of both targeted and control phantom channels for different acoustic conditions listed in Table 2 (G_{A1} and G_{A2}) are illustrated in Figure 5. Each curve represents the average over 10 trials.

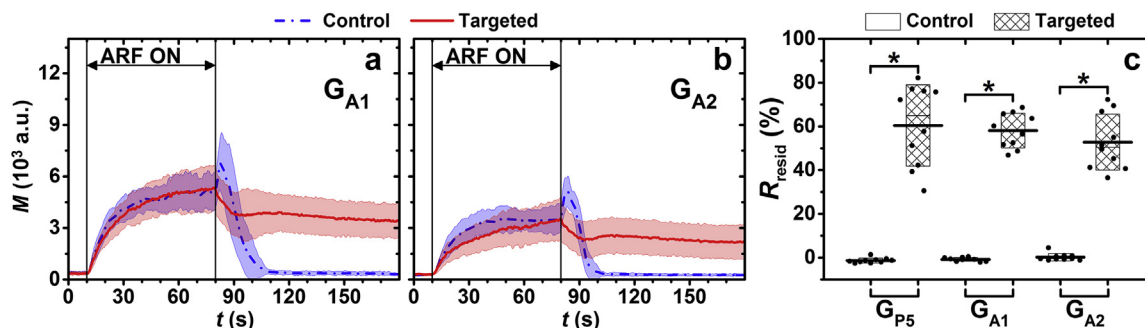


Fig. 5. Signal magnitude curves of adherent microbubbles in control (blue) and targeted (red) phantom channels at different attenuations listed in Table 2. (a) G_{A1} , the attenuation group with one rubber membrane layer (2.1 dB attenuation) and (b) G_{A2} , the attenuation group with two rubber membrane layers (4.4 dB attenuation). Blue dash-dot and red solid lines indicate the mean values from 10 trials. Light color shadows indicate the corresponding error bars at the range of [mean $\pm$ standard deviation]. (c) The R_{resid} parameter for G_{P5} (reference group), G_{A1} , and G_{A2} . Boxes (empty for control and filled for targeted phantom channels) show the range of [mean $\pm$ standard deviation]. Black lines located at the center of the boxes show the corresponding mean value from 10 trials. Raw data from 10 trials are depicted as solid dots overlaid with corresponding boxes. For Student's t -test, *: $p < 0.001$, $n = 10$.

Compared with the reference group (G_{P5}) results depicted in Figure 3e, a higher attenuation resulted in lower signal magnitude for both targeted and control phantom channels. However, the shape of average signal magnitude curves for both targeted and control phantom channels remained similar at different attenuation levels. The R_{resid} parameters of targeted phantom channels were significantly higher than those of control phantom channels ($p < 0.001$, $n = 10$). In addition, there was no significant difference ($p > 0.28$, $n = 10$) in R_{resid} values between any two of the three groups (G_{A1} , G_{A2} , and G_{P5}).

Effects of ARF intensity on R_{resid}

The signal magnitude curves of both targeted and control phantom channels for different time-averaged ARF intensities (G_{PRF1} and G_{PRF2}) are illustrated in Figure 6. Each curve represented the average over 10 trials. Compared with the reference group (G_{P5}) in Figure 3, higher ARF intensities resulted in higher signal magnitude levels for only the targeted phantom channels. For control phantom channels, both the saturation signal magnitude ($t = 80$ s) and the shape of signal magnitude curve remained similar for the three groups (G_{PRF1} , G_{PRF2} , and G_{P5}). The R_{resid} parameters in targeted phantom channels were significantly higher than those of control phantom channels ($p < 0.001$, $n = 10$). In addition, R_{resid} values in the G_{PRF2} group were significantly lower than that of the reference group ($p < 0.05$, $n = 10$).

Correlation between R_{resid} and streptavidin concentration under different flow velocities

Six representative signal magnitude curves from targeted phantom channels with different incubation concentrations of streptavidin and flow velocities, listed in Table 3 ($G_{V1\ C1}$, $G_{V1\ C4}$, $G_{V1\ C6}$, $G_{V2\ C1}$, $G_{V2\ C4}$, and

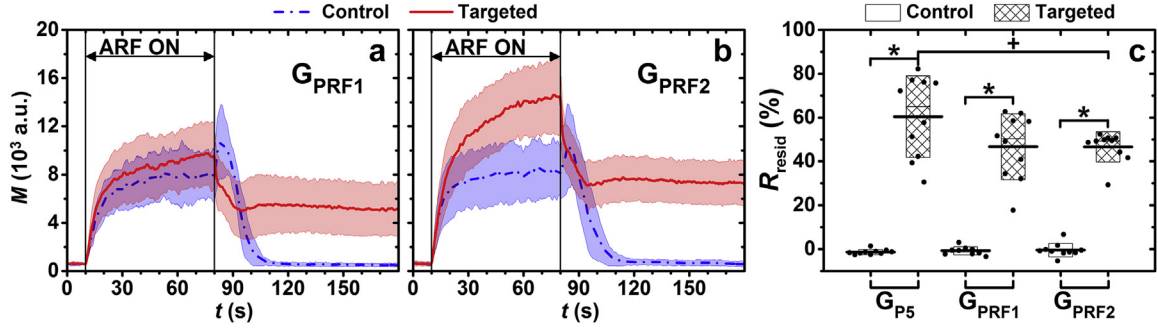


Fig. 6. Signal magnitude curves of adherent microbubbles in control (blue) and targeted (red) phantom channels at different PRF of ARF pulses, (a) G_{PRFF1} and (b) G_{PRFF2} . Blue dash-dot and red solid lines indicate the mean values from 10 trials. Light color shadows indicate the corresponding error bars at the range of [mean $\pm$ standard deviation]. (c) The R_{resid} parameter for groups of G_{P5} (reference group), G_{PRFF1} , and G_{PRFF2} . Boxes (empty for control and filled for targeted phantom channels) show the range of [mean $\pm$ standard deviation]. Black lines located at the center of the boxes show the corresponding mean value from 10 trials. Raw data from 10 trials are shown as solid dots overlaid with corresponding boxes. For Student's t -test, †: $p < 0.05$, *: $p < 0.001$, $n = 10$.

$G_{V2\ C6}$), are illustrated in Figure 7. Each curve represents the average over 10 trials. For each of the two flow velocities (6 and 10 cm/s), higher incubation concentration resulted in higher residual levels. For the two groups with an incubation concentration of 1 $\mu\text{g/mL}$ ($G_{V1\ C1}$ and $G_{V2\ C1}$), the residual signal magnitudes were not significantly higher than the corresponding initial signal magnitudes ($p > 0.16$, $n = 10$). In other groups with the same incubation concentrations but different flow velocity, the higher flow velocity resulted in lower residual levels.

The measured molecular marker (Eu^{3+} -labelled streptavidin) surface concentration (C_{Molecule}) through spectrofluorometry linearly increased ($R^2 = 0.985$) with incubation concentration ($C_{\text{Incubation}}$) (Fig. 8a). There were 5 trials for each of the incubation concentrations. The linear relationship between molecular marker concentration and incubation concentration can be described as:

$$C_{\text{Molecule}} = (6.9 \pm 1.0) \times 10^{-2} C_{\text{Incubation}} \quad (2)$$

The unit of C_{Molecule} and $C_{\text{Incubation}}$ are $10^3/\mu\text{m}^2$ and $\mu\text{g/mL}$, respectively. The residual signal magnitude (M_{resid}) was found to be linear with molecular marker concentration at different flow velocities ($R^2 = 0.985$ for 6 cm/s, $R^2 = 0.983$ for 10 cm/s) (Fig. 8b, d). Additionally, the R_{resid} parameter was also found to be linear with molecular marker concentration at different flow velocities ($R^2 = 0.998$ for 6 cm/s, $R^2 = 0.949$ for 10 cm/s) (Fig. 8c, e).

To further characterize the performance of R_{resid} as a quantitative measure of molecular marker concentration, the detection sensitivity was calculated as the minimum difference in molecular marker concentration that could provide significantly different R_{resid} values. The detection sensitivity was not the same at different molecular marker concentrations: lower molecular marker concentrations were associated with higher detection sensitivity for both flow velocities (6 and 10 cm/s). At the 6 cm/s flow velocity, the highest and lowest detection sensitivities were $0.9 \times 10^3/\mu\text{m}^2$ and $1.3 \times 10^3/\mu\text{m}^2$, respectively (Fig. 8c). At the flow velocity of 10 cm/s, the highest and lowest detection sensitivities were $2.2 \times 10^3/\mu\text{m}^2$ and $2.6 \times 10^3/\mu\text{m}^2$, respectively (Fig. 8e).

DISCUSSION

A summary of tested hypotheses, results, and conclusions from this study are summarized in Figure 9. As illustrated in Figures 3 and 4, groups (G_{P3} , G_{P6} , and G_{P9}) with the highest ARF acoustic pressure (189.5 kPa) resulted in the lowest residual signal magnitudes in targeted phantom channels and relatively lower R_{resid} ($\sim 30\%$) values compared with the other six groups (listed in Table 1). Additionally, targeted channels

Table 3. Parameters of experiments for measurements of molecular marker concentration

Parameter	Group name					
	$G_{V1\ C1}$	$G_{V1\ C2}$	$G_{V1\ C3}$	$G_{V1\ C4}$	$G_{V1\ C5}$	$G_{V1\ C6}$
v^* (cm/s)	6	6	6	6	6	6
$C_{\text{Incubation}}^\dagger$ ($\mu\text{g/mL}$)	1	10	20	30	40	50

Parameter	Group name					
	$G_{V2\ C1}$	$G_{V2\ C2}$	$G_{V2\ C3}$	$G_{V2\ C4}$	$G_{V2\ C5}$	$G_{V2\ C6}$
v (cm/s)	10	10	10	10	10	10
$C_{\text{Incubation}}$ ($\mu\text{g/mL}$)	1	10	20	30	40	50

* Flow velocity.

† Incubation concentration of streptavidin.

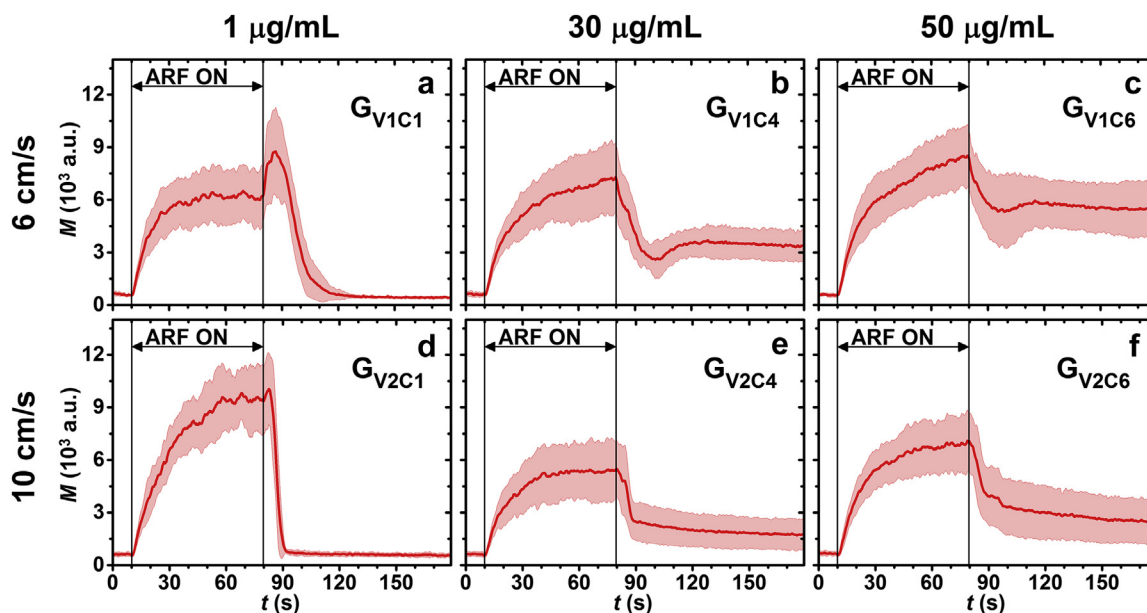


Fig. 7. Signal magnitude curves of adherent microbubbles in targeted phantom channels at different flow and incubation conditions listed in Table 3. (a) G_{V1C1} , (b) G_{V1C4} , (c) G_{V1C6} , (d) G_{V2C1} , (e) G_{V2C4} , (f) G_{V2C6} . Red solid lines indicate the mean values from 10 trials. Light color shadows indicate the corresponding error bars at the range of [mean $\pm$ standard deviation].

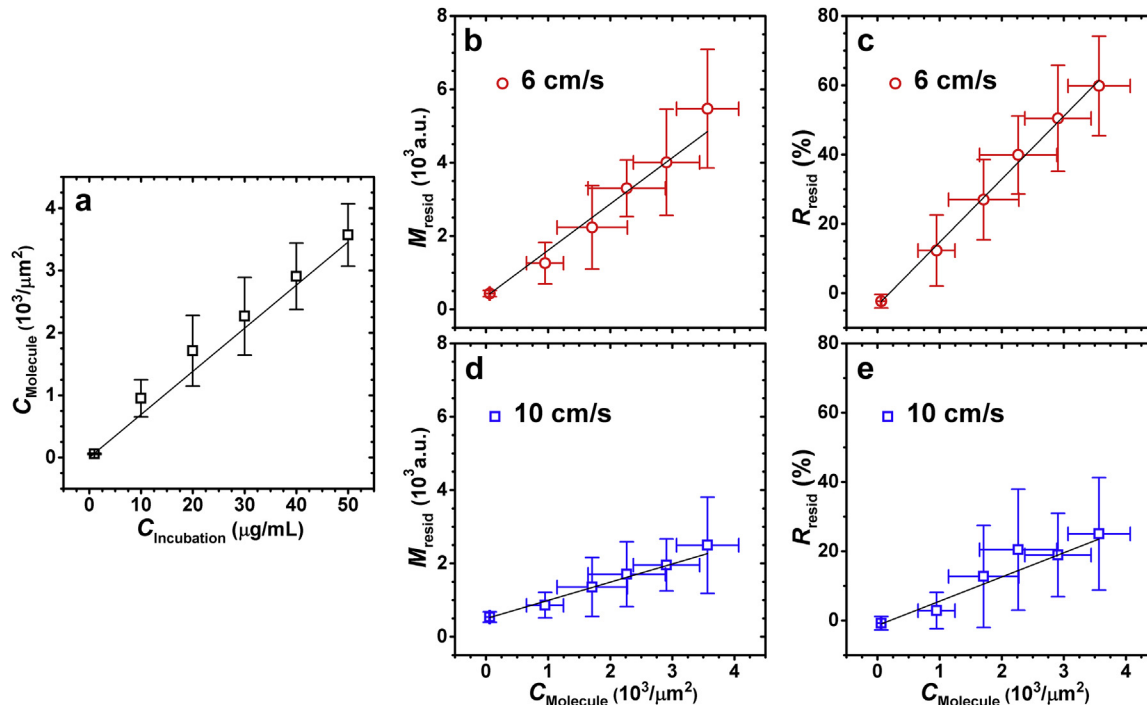


Fig. 8. (a) Relationship between incubation concentration ($C_{Incubation}$) and measured molecular marker concentration ($C_{Molecule}$) of Eu^{3+} -labelled streptavidin ($R^2 = 0.985$). Relationships between residual signal magnitude (M_{resid}) and molecular marker concentration at flow velocities of 6 cm/s ($R^2 = 0.985$) (b), and 10 cm/s ($R^2 = 0.983$) (d). Relationships between R_{resid} parameter and molecular marker concentration at flow velocities of 6 cm/s ($R^2 = 0.998$) (c), and 10 cm/s ($R^2 = 0.949$) (e). Black lines show the weighted linear fits; error bars show the range of [mean $\pm$ standard deviation].

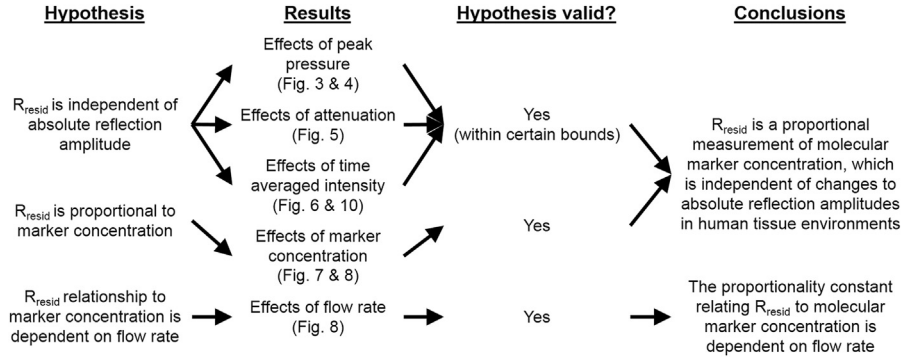


Fig. 9. Summary of tested hypotheses, results, and conclusions.

in the other six groups (G_{P1} , G_{P2} , G_{P4} , G_{P5} , G_{P7} , and G_{P8}) with varied imaging and ARF acoustic pressures exhibited the same relatively high R_{resid} values ($\sim 60\%$). Consequently, below a certain limit in ARF acoustic pressure (between 106.0 and 189.5 kPa in this study), the results indicated that the parameter R_{resid} of targeted channels was independent of both imaging and ARF acoustic pressures. Although a low MI of 0.089 was maintained, which was below the limit at which microbubble rupture is common (Bouakaz *et al.*, 2005), it is hypothesized that the relatively high ARF acoustic pressure (189.5 kPa) destroyed or dislodged adherent microbubbles thus resulting in lower residual signal magnitudes as well as lower R_{resid} values. In addition, as illustrated in Figure 5, all targeted channels under varied levels of attenuation (G_{P5} , G_{A1} , and G_{A2}) exhibited the same R_{resid} values ($\sim 60\%$), which further indicates that the R_{resid} parameter is independent of absolute signal magnitude. Consequently, these results illustrate that when operating below certain acoustic pressure limits, the R_{resid} param-

eter is a measurement that is independent of acoustic pressure and tissue attenuation.

In similar experiments performed with varied levels of time-averaged ARF intensity, instead of peak intensity, it was observed that the higher time-averaged intensity resulted in lower R_{resid} values. Compared with group G_{P5} (Fig. 3e), the targeted signal magnitude curve at highest time-averaged intensity exhibited a sudden magnitude drop immediately after cessation of ARF ($t = 80$ s), which is illustrated by arrows and labels as the difference between magnitude levels M_1 and M_2 in Figure 10(a, b). It is hypothesized that the additional signal magnitude and rapid magnitude drop is due to layering of microbubbles that occurs with increased ARF. This accumulation of microbubbles can result in two groups: (1) a lower sub-layer of microbubbles which directly interacts with the streptavidin coated channel wall; (2) an upper sub-layer of microbubbles which accumulates on top of the lower sub-layer due to higher ARF intensity but is not accessible to the streptavidin along the channel wall.

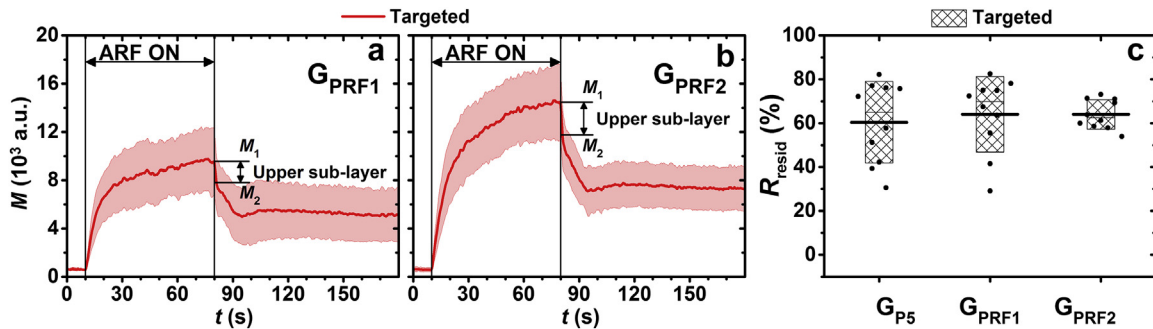


Fig. 10. Signal magnitude curves of adherent microbubbles in targeted phantom channels at different PRF of ARF pulses, (a) G_{PRFF1} and (b) G_{PRFF2} . Solid lines indicate the mean values from 10 trials. Light color shadows indicate the corresponding error bars at the range of $[\text{mean} \pm \text{standard deviation}]$. M_1 is the maximum signal magnitude at $t = 80$ s. M_2 is the signal magnitude at $t = 81$ s. The signal magnitude of upper sub-layer of microbubbles is assumed to be $M_2 - M_1$. (c) The R_{resid} parameter for groups of G_{P5} (reference group), G_{PRFF1} , and G_{PRFF2} . Filled boxes show the range of $[\text{mean} \pm \text{standard deviation}]$. Black lines located at the center of the boxes show the corresponding mean value from 10 trials. Raw data from 10 trials are shown as solid dots overlaid with corresponding boxes.

Therefore, immediately after cessation of ARF the upper sub-layer of microbubbles was rapidly freed from the aggregation because there was no molecular binding force to retain them. In contrast, the microbubbles in the lower sub-layer, which are able to interact with the streptavidin, underwent the normal exponential decay to a steady-state residual level. The rapid dissipation of the former group of microbubbles is therefore hypothesized to have resulted in the sudden magnitude drops uniquely exhibited by groups G_{PRF1} and G_{PRF2} . Consequently, M_1 and M_2 represent the signal magnitudes of the sum of two sub-layers and the lower sub-layer alone, respectively (Fig. 10a, b). If the microbubble layering effect was taken into account by replacing M_1 with M_2 while calculating R_{resid} values using Eqn. 1, there was actually no significant difference of R_{resid} values ($\sim 60\%$) between any two of the three groups (G_{P5} , G_{PRF1} , and G_{PRF2}) with a minimum p value of 0.57 ($n = 10$). Hence, results suggested that parameter R_{resid} was again a robust measurement, which was immune to ARF intensity if the microbubble layering effect is taken into account. Note that in these R_{resid} measurements, it was assumed that the upper sub-layer of microbubbles was completely dissipated within 1 s (from $t = 80$ s to 81 s).

The results in Figure 8a indicate excellent linearity between molecular marker concentration and incubation concentration (Eqn. 2, $R^2 = 0.985$). Previous studies demonstrated that ICAM-1 expression levels for endothelial cells can be as high as approximately 5×10^6 per cell (Almenar-Queralt et al., 1995; Dustin and Springer, 1988). The endothelial cell surface area was approximately $10^3 \mu m^2$ (Adamson, 1993). Hence the site densities of ICAM-1 ranged approximately from less than 50 (unstimulated) to 5×10^3 (stimulated) sites/ μm^2 (Almenar-Queralt et al., 1995; Dustin and Springer, 1988). The experimental results corresponded well with our measured streptavidin concentrations (approximately 0 to $4.0 \times 10^3 \mu m^{-2}$). However, it is anticipated that the measured molecular marker concentration of Eu^{3+} -labelled streptavidin through spectrofluorometry was overestimated in this study. The average pore size in agarose gelatin phantom was approximately 200 nm (Bauman et al., 2004; Griess et al., 1993). Presumably, Eu^{3+} -labelled streptavidin molecules could permeate the phantom through the holes on surface during incubation. As a result, when washing the phantom channels, enhancement solution would take away Eu^{3+} from not only the streptavidin molecules attached onto the channel surface, but also a portion of streptavidin molecules embedded inside the phantom within a certain depth. This portion of streptavidin molecules embedded inside the phantom actually could not participate in the binding process with targeted microbubbles.

The linear regression analysis on the R_{resid} values versus streptavidin concentration illustrated in Figure 8 indicated the feasibility of parameter R_{resid} as a quantitative measurement of molecular marker concentration. However, detection sensitivity decreased with higher flow rates because of the fewer residual microbubbles caused by higher shear forces. Based on the site densities of ICAM-1 (approximately 50 to 5×10^3 sites/ μm^2) (Almenar-Queralt et al., 1995; Dustin and Springer, 1988), the detection sensitivities at flow rates of both 6 and 10 cm/s would be sufficient to distinguish blood vessels with stimulated versus unstimulated endothelial cells.

CONCLUSIONS

Feasibility was illustrated in this study for quantification of molecular marker concentration using a new modulated ARF-based imaging sequence and extraction of the residual-to-saturation ratio (R_{resid}) parameter. Various experimental parameters including acoustic pressures, intensities and attenuation were used to test this imaging strategy. This technique exhibited many desirable characteristics: (1) the quantitative measurement parameter (R_{resid}) is proportional to molecular marker concentration, and within certain bounds, is independent of attenuation and absolute signal magnitude; (2) no separate control groups were required; (3) short imaging protocols were used with scan time of 3 min and no extra waiting period. However, the quantitative measurement was found to be dependent on flow rates, with lower detection sensitivity at higher flow rates. Overall, this study illustrates the feasibility of quantitative molecular marker measurements using non-invasive ultrasound-based targeted molecular imaging. It represents a promising technique with high potential for improved ultrasound molecular imaging for real-time detection and quantification of early disease markers for atherosclerosis in large arteries *in vivo*.

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